Day 6: Lists in Python (Detailed Notes for Data Analysis Students)

What is a List?

A List is a collection of multiple items stored in a single variable.

Lists are:

✅ Ordered

✅ Changeable (Mutable)

✅ Allow Duplicate Values

WHY ARE LISTS IMPORTANT?

In Data Analysis, lists are used to store:

- Employee Names
- Sales Records
- Product Names
- Customer IDs
- Monthly Revenue

Example:

```
sales = [10000, 20000, 30000, 40000]
```

Instead of creating:

```
sales1 = 10000
sales2 = 20000
sales3 = 30000
sales4 = 40000
```

we store everything in one list.

Creating a List

```
names = ["Rahul", "Amit", "Priya"]

print(names)
```

Output:

```
['Rahul', 'Amit', 'Priya']
```

List Can Store Different Data Types

```
data = ["Rahul", 25, 35000.50, True]

print(data)
```

Output:

```
['Rahul', 25, 35000.5, True]
```

List Indexing

Each item has a position.

```
Names List

Rahul   Amit   Priya   Neha

 0       1       2      3
```

Example:

```
names = ["Rahul", "Amit", "Priya"]

print(names[0])
```

Output:

```
Rahul
```

Negative Indexing

```
Rahul   Amit   Priya

 -3      -2      -1
```

```
names = ["Rahul", "Amit", "Priya"]

print(names[-1])
```

Output:

```
Priya
```

List Slicing

Syntax:

```
list[start:end]
```

Example:

```
names = ["Rahul","Amit","Priya","Neha"]

print(names[1:3])
```

Output:

```
['Amit', 'Priya']
```

Changing List Values

```
names = ["Rahul","Amit","Priya"]

names[1] = "Vikas"

print(names)
```

Output:

```
['Rahul', 'Vikas', 'Priya']
```

Add Items

APPEND()

Adds item at the end.

```
names = ["Rahul","Amit"]

names.append("Priya")

print(names)
```

Output:

```
['Rahul', 'Amit', 'Priya']
```

---

insert()

Adds item at specific position.

```
names = ["Rahul","Priya"]

names.insert(1,"Amit")

print(names)
```

Output:

```
['Rahul', 'Amit', 'Priya']
```

Remove Items

REMOVE()

```
names = ["Rahul","Amit","Priya"]

names.remove("Amit")

print(names)
```

Output:

```
['Rahul', 'Priya']
```

POP()

Removes item by index.

```
names = ["Rahul","Amit","Priya"]

names.pop(1)

print(names)
```

Output:

```
['Rahul', 'Priya']
```

Delete Entire List

```
names = ["Rahul","Amit"]

del names
```

List Length

```
names = ["Rahul","Amit","Priya"]

print(len(names))
```

Output:

```
3
```

Sorting Lists

SORT()

```
numbers = [50,20,10,40]

numbers.sort()

print(numbers)
```

Output:

```
[10, 20, 40, 50]
```

Reverse Sorting

```
numbers = [50,20,10,40]

numbers.sort(reverse=True)

print(numbers)
```

Output:

```
[50, 40, 20, 10]
```

Reverse List

```
numbers = [10,20,30]

numbers.reverse()

print(numbers)
```

Output:

```
[30, 20, 10]
```

Loop Through List

```
employees = ["Rahul","Amit","Priya"]

for emp in employees:
    print(emp)
```

Output:

```
Rahul
Amit
Priya
```

Check Item Exists

```
employees = ["Rahul","Amit","Priya"]

print("Rahul" in employees)
```

Output:

```
True
```

List Functions

MAX()

```
sales = [10000,20000,15000]

print(max(sales))
```

Output:

```
20000
```

MIN()

```
sales = [10000,20000,15000]

print(min(sales))
```

Output:

```
10000
```

SUM()

```
sales = [10000,20000,15000]

print(sum(sales))
```

Output:

```
45000
```

Nested Lists

List inside another list.

```
employees = [
    ["Rahul",25000],
    ["Amit",30000],
    ["Priya",40000]
]

print(employees)
```

Output:

```
[['Rahul', 25000], ['Amit', 30000], ['Priya', 40000]]
```

Access Nested List Data

```
employees = [
    ["Rahul",25000],
    ["Amit",30000]
]

print(employees[0][0])
```

Output:

```
Rahul
```

List Comprehension

Short way to create lists.

Traditional:

```
numbers = []

for i in range(1,6):
    numbers.append(i)
```

List Comprehension:

```
numbers = [i for i in range(1,6)]
```

```
print(numbers)
```

Output:

```
[1, 2, 3, 4, 5]
```

Real Data Analyst Examples

EXAMPLE 1: TOTAL SALES

```
sales = [12000,25000,18000,30000]

print(sum(sales))
```

Output:

```
85000
```

EXAMPLE 2: HIGHEST SALES

```
sales = [12000,25000,18000,30000]

print(max(sales))
```

Output:

```
30000
```

EXAMPLE 3: AVERAGE SALES

```
sales = [12000,25000,18000,30000]

avg = sum(sales)/len(sales)

print(avg)
```

Output:

```
21250.0
```

Mini Project 1: Student Marks

```
marks = [80,90,75,85,95]

print("Total =", sum(marks))
print("Highest =", max(marks))
print("Lowest =", min(marks))
```

Mini Project 2: Employee Salary Report

```
salary = [25000,30000,35000]

for s in salary:
    print(s)
```

Mini Project 3: Product List

```
products = []

products.append("Laptop")
products.append("Mobile")
products.append("Keyboard")

print(products)
```

Practice Questions

BASIC

1. Create a list of names.
2. Create a list of cities.
3. Create a list of salaries.
4. Print all items.
5. Find list length.

INDEXING

6. Print first item.
7. Print last item.
8. Print second item.
9. Print third item.

10. Use negative indexing.

SLICING

11. Print first 3 items.
12. Print last 2 items.
13. Print middle items.
14. Reverse using slicing.
15. Extract subset.

ADD & REMOVE

16. Add new employee.
17. Add city.
18. Insert product.
19. Remove item.
20. Delete item using pop().

SORT & REVERSE

21. Sort numbers.
22. Sort names.
23. Reverse list.
24. Sort descending.
25. Sort salaries.

LOOPS

26. Print all employees.
27. Print all sales.
28. Calculate total.
29. Find highest value.
30. Find lowest value.

NESTED LISTS

31. Create employee table.
32. Display employee names.
33. Display salaries.
34. Add employee record.

35. Update salary.

BUSINESS PROBLEMS

36. Monthly sales report.
37. Employee salary report.
38. Product inventory.
39. Customer list.
40. Region-wise sales.

Assignment (Data Analyst Level)

Create a Sales Analysis Program:

```
sales = [12000, 25000, 18000, 30000, 22000]
```

Display:

```
Total Sales    : ?
Average Sales  : ?
Highest Sales  : ?
Lowest Sales   : ?
```

Also print all sales values using a loop.

Interview Questions

1. What is a list?
2. Why are lists used?
3. Difference between list and string?
4. What is indexing?
5. What is slicing?
6. What is append()?
7. What is insert()?
8. Difference between remove() and pop()?
9. What is a nested list?
10. What is list comprehension?

Learning Outcome

After Day 6, you should be able to:

✅ Create and manage lists

✅ Access data using indexing and slicing

✅ Add, update, and remove items

✅ Sort and analyze data

✅ Use loops with lists

✅ Build simple sales and employee analysis programs

DAY 7 TOPICS:

- Tuples
- Sets
- Dictionaries
- Dictionary Methods
- Real Data Analysis Examples
- Business Projects
- 50+ Practice Questions
- Interview Questions for Data Analyst Roles